

This document provides additional assistance with wiring your Extron IP Link Pro Control Processor to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instructions, refer to the user's manual for the specific IP Link Pro Control Processor or the documentation supplied by the manufacturer of the controlled device.

For more information on using Global Scriptor Modules, refer to the "[Guide to Using Scriptor Modules](#)" document.

Device Specifications

Device Type: Video Projector
 Manufacturer: Epson
 Firmware Version: Main: 97010166G7WWA093
 HDBaseT: V40701D0600000101
 Pixel Shift: 8YF1WWH001
 Document Camera: ----
 HDBaseT Tx: VFFFFFFFFFFFFFFFFF
 Model(s): EB-800F, EB-805F, PowerLite 800F, PowerLite 805F, CB-800F, CB-805F

Tested on the Following Software and Firmware Versions

IP Link Pro Control Processor Firmware	Global Scriptor Version
3.15.0000-b004	2.14.0

Version History

Module Version	Date	Notes
1_0_0_0	7/12/2022	Initial Version

Module Notes

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'.
Example: `InterfaceName.Unidirectional = 'True'`
- connectionCounter variable must be set to the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15.
Example: `InterfaceName.connectionCounter = 5`
- It is recommended to have a command pacing of 0.1 seconds between every command.
- Please use a 30 second command delay when Power On is used.

Supported Classes and Examples

SerialClass
<code>InterfaceName = ModuleName.SerialClass(ProcessorName, 'COM1', Model='EB-800F')</code>
SerialOverEthernetClass
<code>InterfaceName = ModuleName.SerialOverEthernetClass('192.168.254.254', 2001, Model='EB-800F')</code>
EthernetClass
<code>InterfaceName = ModuleName.EthernetClass('192.168.254.254', 3629, Model='EB-800F')</code>

Control Commands

Format with Qualifier:

```
InterfaceName.Set(Command, Value, {'Qualifier Key': 'Qualifier Value'})
```

Format without Qualifier:

```
InterfaceName.Set(Command, Value)
```

Command AspectRatio ^{1, 2, 5}	Value 'Auto' 'Native'	Value 'Full'	Value 'Zoom'
# AspectRatio example InterfaceName.Set('AspectRatio', 'Auto')			
Command AutoImage ²	Value None		
# AutoImage example InterfaceName.Set('AutoImage', None)			
Command AVMute	Value 'On'	Value 'Off'	
# AVMute example InterfaceName.Set('AVMute', 'On')			
Command ClosedCaption	Value 'Off'	Value 'CC1'	Value 'CC2'
# ClosedCaption example InterfaceName.Set('ClosedCaption', 'Off')			
Command Freeze	Value 'On'	Value 'Off'	
# Freeze example InterfaceName.Set('Freeze', 'On')			
Command Input ⁵	Value 'HDMI 1' 'Computer 1' 'Screen Mirroring' 'Video (RCA)'	Value 'HDMI 2' 'Computer 2' ³ 'HDBaseT' 'USB 1'	Value 'HDMI 3' 'LAN' 'Video' 'USB 2'
# Input example InterfaceName.Set('Input', 'HDMI 1')			
Command LampMode	Value 'Normal' 'Custom'	Value 'Quiet'	Value 'Extended'
# LampMode example InterfaceName.Set('LampMode', 'Normal')			
Command MenuNavigation	Value 'Menu' 'Left' 'Esc'	Value 'Up' 'Right'	Value 'Down' 'Enter'
# MenuNavigation example InterfaceName.Set('MenuNavigation', 'Menu')			
Command Power	Value 'On'	Value 'Off'	
# Power example InterfaceName.Set('Power', 'On')			

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Command SplitScreen	Value 'On'	Value 'Off'	
# SplitScreen example InterfaceName.Set('SplitScreen', 'On')			
Command SplitScreenSize ⁶	Value 'Equal'	Value 'Larger Left'	Value 'Larger Right'
# SplitScreenSize example InterfaceName.Set('SplitScreenSize', 'Equal')			
Command SplitScreenSource ^{4,6}	Value 'HDMI 1' 'Computer 1' 'Screen Mirroring' 'USB 1'	Value 'HDMI 2' 'Computer 2' ³ 'HDBaseT' 'USB 2'	Value 'HDMI 3' 'LAN' 'Video (RCA)'
Qualifier Key 'Side'	Qualifier Value 'Left'	Qualifier Value 'Right'	
# SplitScreenSource example InterfaceName.Set('SplitScreenSource', 'HDMI 1', {'Side': 'Left'})			
Command Volume	Value 0 to 20 in steps of 1		
# Volume example InterfaceName.Set('Volume', 20)			

¹ Availability depends on the Menu → Installation → Screen Type setting.

² Availability depends on the current Input.

³ Ensure that Menu → Signal I/O → Monitor Out Port is set to Computer2.

⁴ Source combinations are limited. Please see the menu for available options when Split Screen is On.

⁵ Unavailable when Split Screen is On.

⁶ Only available when Split Screen is On.

Status Available

For all commands, call Update to receive the latest status. ConnectionStatus does not support the Update function and is triggered by the device providing a successful response to other Update function calls.

Format with Qualifier:

```
InterfaceName.Update(Command, {'Qualifier Key': 'Qualifier Value'})
Value = InterfaceName.ReadStatus(Command, {'Qualifier Key': 'Qualifier Value'})
InterfaceName.SubscribeStatus(Command, {'Qualifier Key': 'Qualifier Value'},
```

FeedbackHandler)

FeedbackHandler will be called only when the specified qualifier gets a new status.

Format without Qualifier:

```
InterfaceName.Update(Command)
Value = InterfaceName.ReadStatus(Command)
InterfaceName.SubscribeStatus(Command, None, FeedbackHandler)
FeedbackHandler will be called when any qualifier gets a new status.
```

Command AspectRatio ^{1,2}	Value 'Auto' 'Native'	Value 'Full'	Value 'Zoom'
# AspectRatio example InterfaceName.Update('AspectRatio') Value = InterfaceName.ReadStatus('AspectRatio') InterfaceName.SubscribeStatus('AspectRatio', None, FeedbackHandler)			
Command AVMute	Value 'On'	Value 'Off'	
# AVMute example InterfaceName.Update('AVMute') Value = InterfaceName.ReadStatus('AVMute') InterfaceName.SubscribeStatus('AVMute', None, FeedbackHandler)			
Command ClosedCaption	Value 'Off'	Value 'CC1'	Value 'CC2'
# ClosedCaption example InterfaceName.Update('ClosedCaption') Value = InterfaceName.ReadStatus('ClosedCaption') InterfaceName.SubscribeStatus('ClosedCaption', None, FeedbackHandler)			
Command ConnectionStatus	Value 'Connected'	Value 'Disconnected'	
# ConnectionStatus example Value = InterfaceName.ReadStatus('ConnectionStatus') InterfaceName.SubscribeStatus('ConnectionStatus', None, FeedbackHandler)			
Command DeviceStatus ⁴	Value 'Normal' 'High Internal Temperature Error' 'Cinema Filter Error' 'Subsystem Error' 'Power Supply Unit Error (Ballast)' 'Cooling System Error'	Value 'Fan Error' 'Lamp Error' 'Electric Dual-Layered Capacitor is Disconnected' 'Low Air Flow Error' 'Shutter Error' 'Static Iris Error'	Value 'Lamp Failure at Power On' 'Open Lamp Cover Door Error' 'Auto Iris Error' 'Air Filter Air Flow Sensor Error' 'Cooling System Error (Peltiert Element)' 'Power Supply Unit Error'

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	(Pump)'		(Disagreement of Ballast)'
	'Exhaust Shutter Error'	'Obstacle Detection Error'	'IF Board Discernment Error'
	'Communication Error of "Stack Projection Function"'		
<pre># DeviceStatus example InterfaceName.Update('DeviceStatus') Value = InterfaceName.ReadStatus('DeviceStatus') InterfaceName.SubscribeStatus('DeviceStatus', None, FeedbackHandler)</pre>			
Command	Value	Value	
Freeze	'On'	'Off'	
<pre># Freeze example InterfaceName.Update('Freeze') Value = InterfaceName.ReadStatus('Freeze') InterfaceName.SubscribeStatus('Freeze', None, FeedbackHandler)</pre>			
Command	Value	Value	Value
Input ⁵	'HDMI 1'	'HDMI 2'	'HDMI 3'
	'Computer 1'	'Computer 2' ³	'LAN'
	'Screen Mirroring'	'HDBaseT'	'Video'
	'Video (RCA)'	'USB 1'	'USB 2'
<pre># Input example InterfaceName.Update('Input') Value = InterfaceName.ReadStatus('Input') InterfaceName.SubscribeStatus('Input', None, FeedbackHandler)</pre>			
Command	Value	Value	Value
LampMode	'Normal'	'Quiet'	'Extended'
	'Custom'		
<pre># LampMode example InterfaceName.Update('LampMode') Value = InterfaceName.ReadStatus('LampMode') InterfaceName.SubscribeStatus('LampMode', None, FeedbackHandler)</pre>			
Command	Value		
LampUsage	Hours		
<pre># LampUsage example InterfaceName.Update('LampUsage') Value = InterfaceName.ReadStatus('LampUsage') InterfaceName.SubscribeStatus('LampUsage', None, FeedbackHandler)</pre>			
Command	Value		
OperationHours	Hours		
<pre># OperationHours example InterfaceName.Update('OperationHours') Value = InterfaceName.ReadStatus('OperationHours') InterfaceName.SubscribeStatus('OperationHours', None, FeedbackHandler)</pre>			
Command	Value	Value	Value
Power	'On'	'Off'	'Warming Up'
	'Cooling Down'		
<pre># Power example InterfaceName.Update('Power') Value = InterfaceName.ReadStatus('Power') InterfaceName.SubscribeStatus('Power', None, FeedbackHandler)</pre>			

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Command	Value
Volume	0 to 20 in steps of 1
<pre># Volume example InterfaceName.Update('Volume') Value = InterfaceName.ReadStatus('Volume') InterfaceName.SubscribeStatus('Volume', None, FeedbackHandler)</pre>	

¹ Availability depends on the Menu → Installation → Screen Type setting.

² Availability depends on the current Input.

³ Ensure that Menu → Signal I/O → Monitor Out Port is set to Computer2.

⁴ Only available for serial control.

⁵ When Split Screen is On, status will display the left input.

Cable and Adapter Requirements

Captive Screw to Female DB9 RS-232 Serial Cable

Notes for the Device

To ensure the functionality of the module, please set the following menu options and restart the projector:

- Menu → Signal I/O → HDBaseT → Control Communications → Off
- Menu → Signal I/O → HDBaseT → Extron XTP → Off

Serial communication

Port Type: RS-232

Baud Rate: 9600

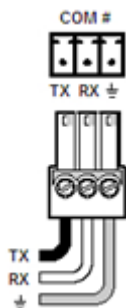
Data Bits: 8

Parity: None

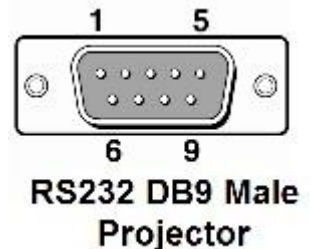
Stop Bits: One

Flow Control: None

Pin Assignments Diagram



Signal	Main Cable	Pin	Signal
TxD	→	2	RxD
RxD	←	3	TxD
GND	→	5	GND



Network communication

When configuring the Ethernet module, be sure device settings match those of the Global Scriptor ethernet interface.

Port Type:	Ethernet
Default Port:	3629
Logon Credentials Supported:	No
Multi-Connection Capabilities:	Yes
Port Changeability:	No

Ethernet Module Configuration Description

Please refer to user manual for settings and changes to the network communication

Notes for the Device

To ensure the functionality of the module, please set the following menu options and restart the projector:

- Menu → Signal I/O → HDBaseT → Control Communications → Off
- Menu → Signal I/O → HDBaseT → Extron XTP → Off
- Menu → Operation *or* ECO → Standby Mode → Communication On

Depending on your device, the following settings may also apply:

- Menu → Operation → Port → Wired LAN *or* Wireless LAN
- Menu → Management → Wireless LAN Power → On
- Menu → Network → Network Settings → Priority Control Interface → Wired LAN *or* Wireless LAN

Appendix A. Set Commands

Aspect Ratio Auto	ASPECT 30\x0D
Aspect Ratio Full	ASPECT 40\x0D
Aspect Ratio Native	ASPECT 60\x0D
Aspect Ratio Zoom	ASPECT 50\x0D
Auto Image None	KEY 4A\x0D
AV Mute Off	MUTE OFF\x0D
AV Mute On	MUTE ON\x0D
Closed Caption CC1	CCAP 11\x0D
Closed Caption CC2	CCAP 12\x0D
Closed Caption Off	CCAP 00\x0D
Freeze Off	FREEZE OFF\x0D
Freeze On	FREEZE ON\x0D
Input Computer 1	SOURCE 10\x0D
Input Computer 2	SOURCE 20\x0D
Input HDBaseT	SOURCE 80\x0D
Input HDMI 1	SOURCE 30\x0D
Input HDMI 2	SOURCE A0\x0D
Input HDMI 3	SOURCE C0\x0D
Input LAN	SOURCE 53\x0D
Input Screen Mirroring	SOURCE 56\x0D
Input USB 1	SOURCE 52\x0D
Input USB 2	SOURCE 54\x0D
Input Video	SOURCE 40\x0D
Input Video (RCA)	SOURCE 41\x0D
Lamp Mode Custom	LUMINANCE 05\x0D
Lamp Mode Extended	LUMINANCE 04\x0D
Lamp Mode Normal	LUMINANCE 00\x0D
Lamp Mode Quiet	LUMINANCE 01\x0D
Menu Navigation Down	KEY 36\x0D
Menu Navigation Enter	KEY 16\x0D
Menu Navigation Esc	KEY 05\x0D
Menu Navigation Left	KEY 37\x0D
Menu Navigation Menu	KEY 03\x0D
Menu Navigation Right	KEY 38\x0D
Menu Navigation Up	KEY 35\x0D
Power Off	PWR OFF\x0D
Power On	PWR ON\x0D
Split Screen Off	SPS 01 00\x0D
Split Screen On	SPS 01 01\x0D
Split Screen Size Equal	SPS 02 00\x0D
Split Screen Size Larger Left	SPS 02 01\x0D
Split Screen Size Larger Right	SPS 02 02\x0D

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Split Screen Source Computer 1 Side Left	SPS 03 10\x0D
Split Screen Source Computer 1 Side Right	SPS 04 10\x0D
Split Screen Source Computer 2 Side Left	SPS 03 20\x0D
Split Screen Source Computer 2 Side Right	SPS 04 20\x0D
Split Screen Source HDBaseT Side Left	SPS 03 80\x0D
Split Screen Source HDBaseT Side Right	SPS 04 80\x0D
Split Screen Source HDMI 1 Side Left	SPS 03 30\x0D
Split Screen Source HDMI 1 Side Right	SPS 04 30\x0D
Split Screen Source HDMI 2 Side Left	SPS 03 A0\x0D
Split Screen Source HDMI 2 Side Right	SPS 04 A0\x0D
Split Screen Source HDMI 3 Side Left	SPS 03 C0\x0D
Split Screen Source HDMI 3 Side Right	SPS 04 C0\x0D
Split Screen Source LAN Side Left	SPS 03 53\x0D
Split Screen Source LAN Side Right	SPS 04 53\x0D
Split Screen Source Screen Mirroring Side Left	SPS 03 56\x0D
Split Screen Source Screen Mirroring Side Right	SPS 04 56\x0D
Split Screen Source USB 1 Side Left	SPS 03 52\x0D
Split Screen Source USB 1 Side Right	SPS 04 52\x0D
Split Screen Source USB 2 Side Left	SPS 03 54\x0D
Split Screen Source USB 2 Side Right	SPS 04 54\x0D
Split Screen Source Video (RCA) Side Left	SPS 03 41\x0D
Split Screen Source Video (RCA) Side Right	SPS 04 41\x0D
Volume 0	VOL 00\x0D
Volume 20	VOL 243\x0D

Appendix B. Update Commands

Aspect Ratio	ASPECT?\x0D
AV Mute	MUTE?\x0D
Closed Caption	CCAP?\x0D
Device Status	ERR?\x0D
Freeze	FREEZE?\x0D
Input	SOURCE?\x0D
Lamp Mode	LUMINANCE?\x0D
Lamp Usage	LAMP?\x0D
Operation Hours	ONTIME?\x0D
Power	PWR?\x0D
Volume	VOL?\x0D